1A. Initial data exploration

1. The attribute type of each attribute and reason

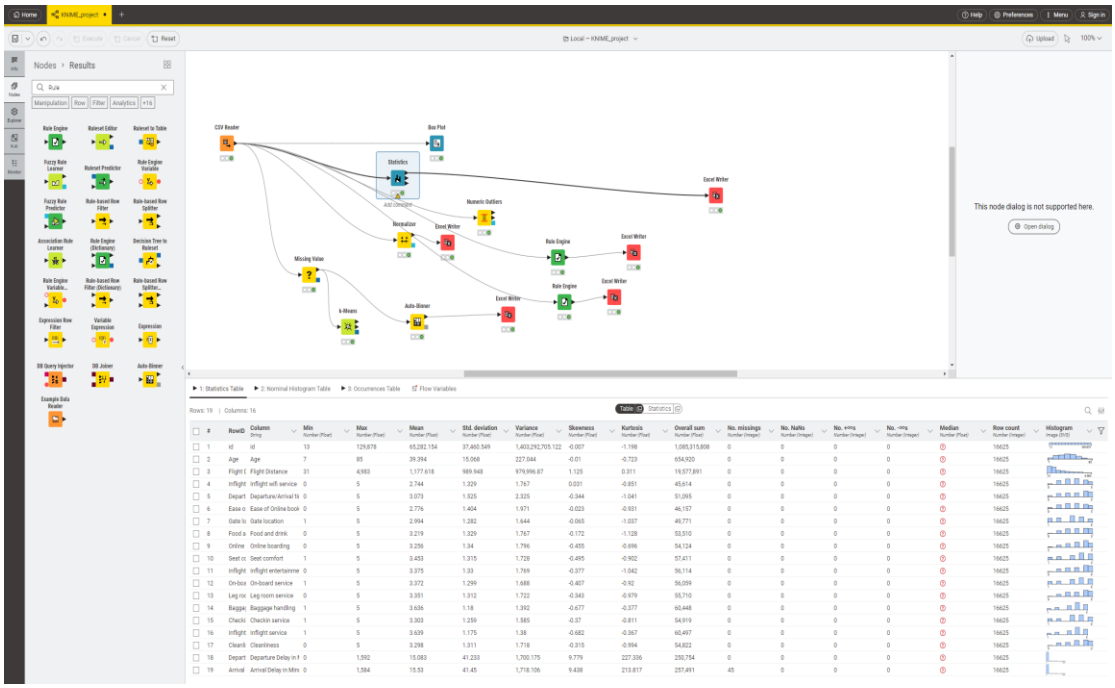
Attribute	Type	Note
id	Nominal	No quantitative meaning, no order
Gender	Nominal	No quantitative meaning, no order
Customer Type	Nominal	No quantitative meaning, no order
Age	Ratio	Has zero and can be calculated
Type of Travel	Nominal	No quantitative meaning, no order
Class	Ordinal	Business > Eco Plus > Eco, but it cannot be calculated
Flight Distance	Ratio	Has zero and can be calculated
Inflight wifi service	Ordinal	Unable to measure the difference between 1 and 5, but 1 to 5 are in order)
Departure/Arrival time convenient	Ordinal	Unable to measure the difference between 1 and 5, but 1 to 5 are in order)
Ease of Online booking	Ordinal	Unable to measure the difference between 1 and 5, but 1 to 5 are in order)
Gate location	Ordinal	Unable to measure the difference between 1 and 5, but 1 to 5 are in order)
Food and drink	Ordinal	Unable to measure the difference between 1 and 5, but 1 to 5 are in order)
Online boarding	Ordinal	Unable to measure the difference between 1 and 5, but 1 to 5 are in order)
Seat comfort	Ordinal	Unable to measure the difference between 1 and 5, but 1 to 11 are in order)
Inflight entertainment	Ordinal	Unable to measure the difference between 1 and 5, but 1 to 5 are in order)

On-board service	Ordinal	Unable to measure the difference between 1 and 5, but 1 to 5 are in order)
Leg room service	Ordinal	Unable to measure the difference between 1 and 5, but 1 to 5 are in order)
Baggage handling	Ordinal	Unable to measure the difference between 1 and 5, but 1 to 5 are in order)
Checkin service	Ordinal	Unable to measure the difference between 1 and 5, but 1 to 5 are in order)
Inflight service	Ordinal	Unable to measure the difference between 1 and 5, but 1 to 5 are in order)
Cleanliness	Ordinal	Unable to measure the difference between 1 and 5, but 1 to 5 are in order)
Departure Delay in Minutes	Ratio	Has zero and can be calculated (Time)
Arrival Delay in Minutes	Ratio	Has zero and can be calculated (Time)
satisfaction	Nominal	No quantitative meaning, no order

2. Attribute statistics

   Example of setting in Knime:

I use the CSV reader node and then connect to the statistics node to get relevant data.



Column	Min	Max	Mean	Std. deviation	Variance	Skewness	Kurtosis
id	13	129878	65282.15386	37460.54865	1403292705	-0.007137641	-1.197795621
Age	7	85	39.39368421	15.06798504	227.0441733	-0.009730498	-0.723383748
Flight Dista	31	4983	1177.617504	989.9479129	979996.8702	1.124724743	0.311255032
Inflight wifi	0	5	2.743699248	1.329246932	1.766897407	0.030989317	-0.850731956
Departure/	0	5	3.073383459	1.524867501	2.325220896	-0.344312738	-1.041371057
Ease of On	0	5	2.776360902	1.403893207	1.970916135	-0.023346715	-0.931387585
Gate locati	1	5	2.993744361	1.282032996	1.643608603	-0.065073704	-1.037002194
Food and	0	5	3.218646617	1.329231479	1.766856325	-0.172171291	-1.127712273
Online boa	0	5	3.255578947	1.340022236	1.795659592	-0.454653682	-0.695537209
Seat comfo	1	5	3.453293233	1.314619026	1.728223183	-0.49532784	-0.902193781
Inflight ent	0	5	3.375278195	1.329987466	1.768866659	-0.377490443	-1.04161393
On-board	1	5	3.371969925	1.299379294	1.688386549	-0.407080284	-0.920078913
Leg room	0	5	3.350977444	1.312307702	1.722151505	-0.342979606	-0.979329382
Baggage h	1	5	3.635969925	1.180041607	1.392498194	-0.676933207	-0.376977357
Checkin se	1	5	3.303398496	1.259078151	1.58527779	-0.36974906	-0.810568057
Inflight ser	1	5	3.638917293	1.174946513	1.380499309	-0.682034836	-0.367075202
Cleanliness	0	5	3.29756391	1.310654297	1.717814686	-0.315403206	-0.993915666
Departure	0	1592	15.08294737	41.2331838	1700.175446	9.778701718	227.3359708
Arrival Del	0	1584	15.53021713	41.45004403	1718.10615	9.438491098	213.8170341

Overall sum	No. missings	No. NaNs	No. +∞s	No. -∞s	Median	Row count
1085315808	0	0	0	0		16625
654920	0	0	0	0		16625
19577891	0	0	0	0		16625
45614	0	0	0	0		16625
51095	0	0	0	0		16625
46157	0	0	0	0		16625
49771	0	0	0	0		16625
53510	0	0	0	0		16625
54124	0	0	0	0		16625
57411	0	0	0	0		16625
56114	0	0	0	0		16625
56059	0	0	0	0		16625
55710	0	0	0	0		16625
60448	0	0	0	0		16625
54919	0	0	0	0		16625
60497	0	0	0	0		16625
54822	0	0	0	0		16625
250754	0	0	0	0		16625
257491	45	0	0	0		16625

Analysis using the KNIME Statistics node reveals some interesting observations. First, the passenger age distribution ranges from 7 to 85 years old, with an average of approximately 39 years old, indicating a broad user base for aviation services. Second, the flight distance distribution exhibits a significant right-skewed characteristic, with the majority of flights being short-haul, but a few long-haul flights significantly exceeding the maximum value. In terms of service ratings, inflight Wi-Fi received the lowest score (2.74/5), while baggage handling and inflight service performed the best (3.63/5), indicating that passengers are highly satisfied with basic services and have higher expectations for digital services (such as Wi-Fi). Finally, the delay data exhibits an extremely skewed distribution: while the vast majority of flights are on time or slightly delayed, a few experience delays exceeding 1,500 minutes. These outliers appear as significant outliers in the data visualization and deserve special attention.

3. Explore multiple attributes

3.1 Clustering research

First, use four rule engine nodes to convert the satisfaction, type of travel, customer type, and gender data into numbers represented by 1 and 0, convert the class column to 0, 1, 2. Then use String to Number to convert

them into a usable table.

K-mean (Satisfaction and Class)

▶ 1: Labeled input ▶ 2: Clusters ■ 3: PMML Cluster Model Flow Variables

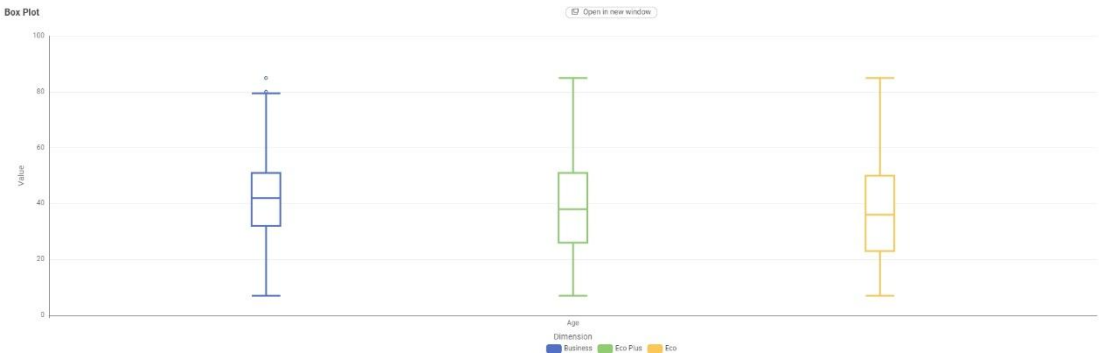
Rows: 3 | Columns: 2 Table Statistics

#	RowID	Age Number (Float)	satisfaction Number (Float)
1	cluster_	40.019	0.497
2	cluster_	57.262	0.486
3	cluster_	21.578	0.303

Using K-Means clustering (K=3), passengers were divided into three groups: young passengers (average age 22, only 30% satisfaction), middle-aged passengers (average age 40, approximately 50% satisfaction), and senior passengers (average age 57, approximately 49% satisfaction). The results showed that younger passengers were significantly less satisfied than the middle-aged and senior groups, likely due to their higher expectations for in-flight entertainment, Wi-Fi, and other features.

3.2 multi-data process

Box Plot (Age and Class)



The chart shows that Business class (blue) has a higher median age (approximately over 40 years old), with a wider age distribution, ranging from their 20s to their 70s, with some older passengers (outliers in their 80s).

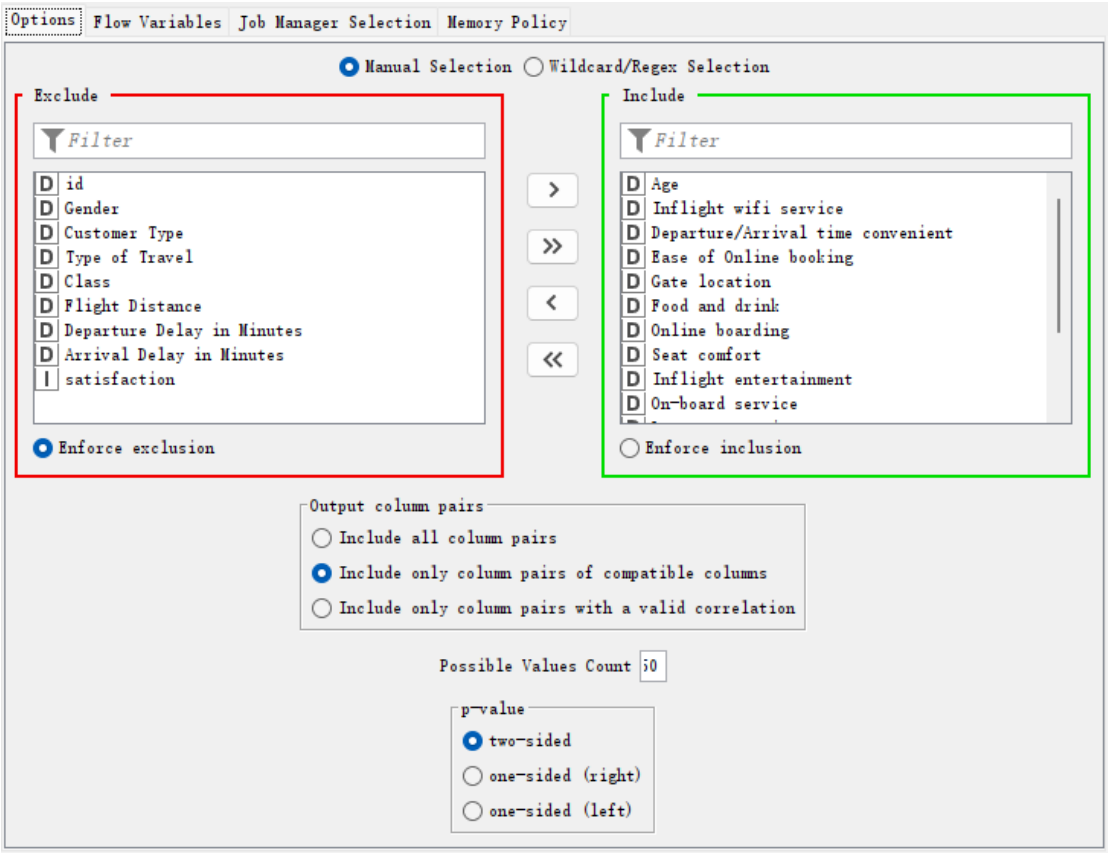
Eco Plus (green) has a slightly lower median age than Business, ranging from approximately 38 to 40 years old. This broad distribution shows a wide age range.

Eco class (yellow) has the lowest median age, around 35 years old. The age distribution skews towards the younger age group, with more passengers in

their 20s and 30s, but there are still some older passengers.

Linear Correlation (Age and service)

To explore the relationship between age and passengers' ratings of various services, I used the Linear Correlation node in KNIME. In the configuration, I included Age and all columns related to the service experience (such as Inflight Wi-Fi service, Seat comfort, Online boarding, and Food and drink) in the Include field, while excluding other irrelevant variables (such as ID, gender, and delay time). The output is a Pearson correlation coefficient matrix, with values ranging from -1 to 1, with values closer to ±1 indicating stronger correlations.



#	RowID	Age Number (Float)	Inflight wifi ser... Number (Float)	Departure/Arri... Number (Float)	Ease of Online ... Number (Float)	Gate location Number (Float)	Food and drink Number (Float)	Online boarding Number (Float)
1	Age	1	0.023	0.046	0.032	-0.004	0.013	0.211

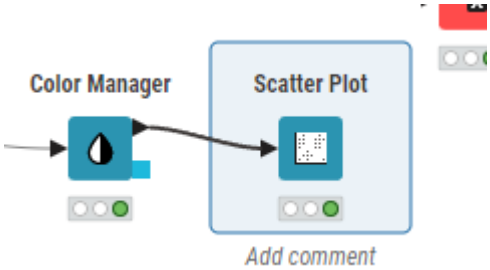
Seat comfort Number (Float)	Inflight enterta... Number (Float)	On-board servi... Number (Float)	Leg room servi... Number (Float)	Baggage handl... Number (Float)	Checkin service Number (Float)	Inflight service Number (Float)	Cleanliness Number (Float)
0.167	0.071	0.061	0.045	-0.051	0.038	-0.055	0.045

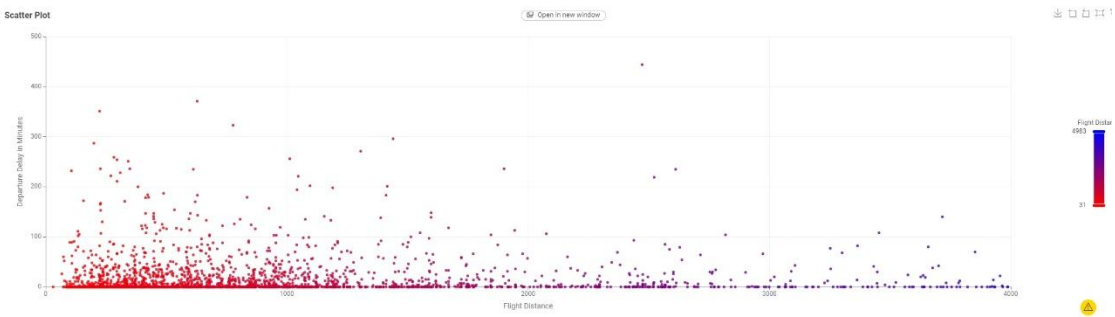
Comprehensive analysis reveals that passenger age has a weak correlation with most service ratings, with only slight positive correlations for online

boarding and seat comfort. This suggests that service satisfaction is largely unrelated to age and is more influenced by other factors, such as flight delays and the overall travel experience.

Scatter Plot (Flight distance and Departure Delay in Minutes)

To study the relationship between flight distance and delay time, I used the Color Manager and Scatter Plot nodes in KNIME. In the Color Manager, I set Flight Distance to a range and mapped it to a gradient from red (short-distance) to blue (long-distance). In the Scatter Plot, I plotted Flight Distance on the X-axis and Departure Delay in Minutes on the Y-axis, with the color of the points determined by Flight Distance.



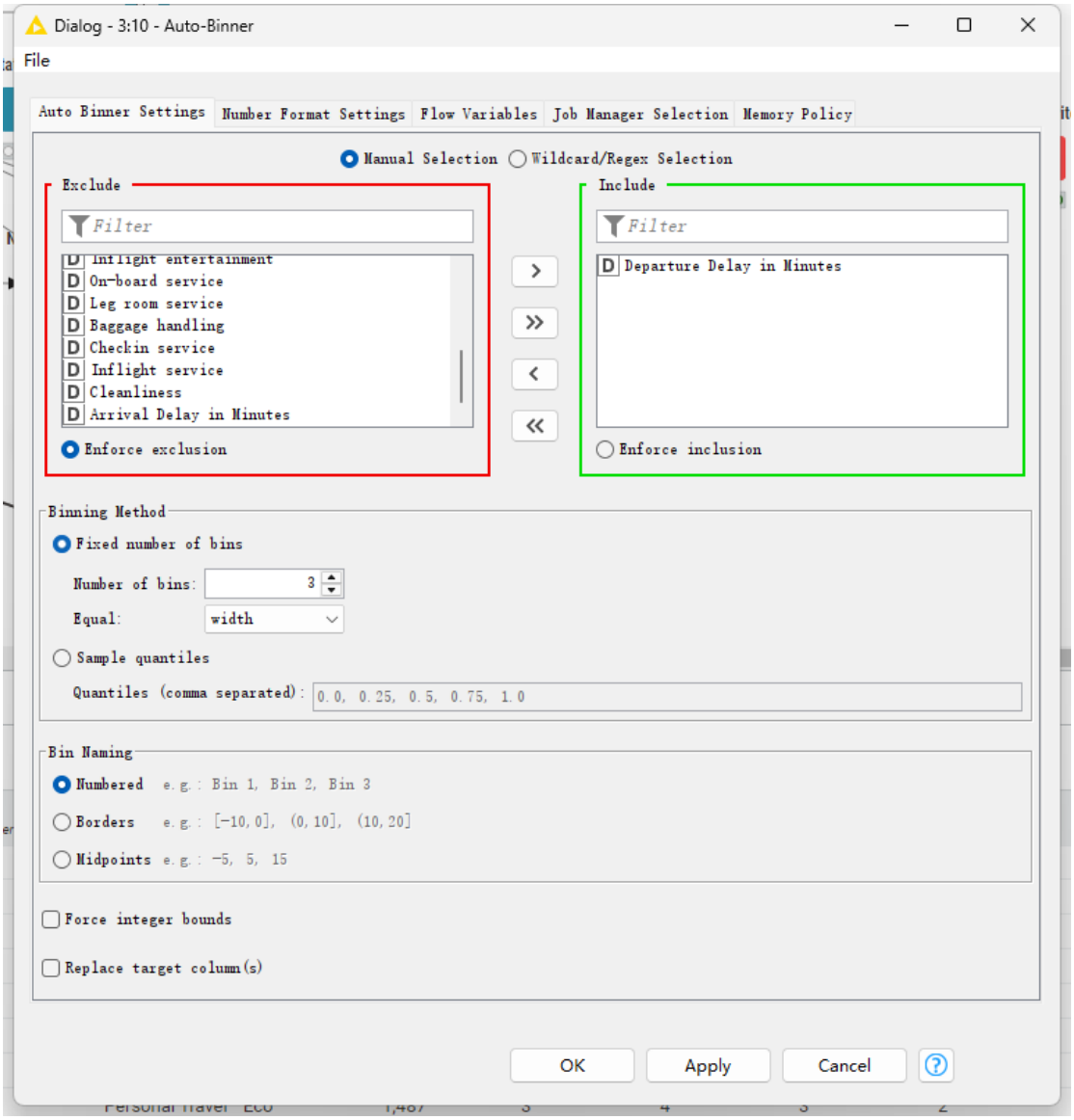


The scatter plot shows that most flights are short-haul (less than 1,000 km), with generally low delays (0–100 minutes) and a reddish tint. Long-haul flights (3,000+ km) are relatively rare, but experience greater fluctuations in delays, with some exceeding 300 minutes, and are colored blue. In summary, there is a relationship between flight distance and flight delays: short-haul flights are more numerous and experience relatively concentrated delays; long-haul flights are fewer in number but are more prone to significant delays. This finding suggests that airlines need to implement stricter delay management measures on long-haul flights.

1B. Data preprocessing

1. Equi-width binning and Equi-depth binning

   Example of binning setting in Knime:



   Departure Delay in Minutes Equi-width:

id	Minutes	Arrival Del	Departure Delay in Minutes [Binned]
49833	0	0	Bin 1
22532	0	0	Bin 1
111275	0	5	Bin 1
74985	0	0	Bin 1
75332	0	7	Bin 1
106685	0	5	Bin 1
86460	0	0	Bin 1
81846	30	24	Bin 1
102610	0	5	Bin 1
114955	0	0	Bin 1
105654	38	28	Bin 1
101667	13	43	Bin 1
49295	0	0	Bin 1
85252	0	0	Bin 1
118594	35	30	Bin 1
114930	1	0	Bin 1
117832	13	10	Bin 1
8915	16	14	Bin 1
74178	73	77	Bin 1
62993	1	0	Bin 1
64239	13	22	Bin 1
22078	41	39	Bin 1
95784	0	13	Bin 1
84358	0	0	Bin 1
89346	0	0	Bin 1
125131	17	13	Bin 1
38574	2	15	Bin 1
43069	60	54	Bin 1
7448	0	0	Bin 1
46013	0	4	Bin 1
39795	0	0	Bin 1
4131	0	0	Bin 1
111519	24	7	Bin 1
89671	0	0	Bin 1
121716	0	0	Bin 1
74329	1	0	Bin 1
24876	0	11	Bin 1
95149	0	0	Bin 1
8511	29	3	Bin 1
119515	0	8	Bin 1
89965	271	275	Bin 1
108035	0	0	Bin 1
52931	8	0	Bin 1
66290	1	0	Bin 1
61862	0	0	Bin 1
73090	0	0	Bin 1
118191	47	35	Bin 1
118825	15	0	Bin 1
13295	0	0	Bin 1
19981	30	22	Bin 1
21992	0	0	Bin 1
101236	4	0	Bin 1
17707	0	0	Bin 1
125570	3	0	Bin 1

Departure Delay in Minutes Equi-depth:

id	Departure	Arrival Del	Departure D
49833	0	0	Bin 1
22532	0	0	Bin 1
111275	0	5	Bin 1
74985	0	0	Bin 1
75332	0	7	Bin 1
106685	0	5	Bin 1
86460	0	0	Bin 1
81846	30	24	Bin 3
102610	0	5	Bin 1
114955	0	0	Bin 1
105654	38	28	Bin 3
101667	13	43	Bin 2
49295	0	0	Bin 1
85252	0	0	Bin 1
118594	35	30	Bin 3
114930	1	0	Bin 2
117832	13	10	Bin 2
8915	16	14	Bin 3
74178	73	77	Bin 3
62993	1	0	Bin 2
64239	13	22	Bin 2
22078	41	39	Bin 3
95784	0	13	Bin 1
84358	0	0	Bin 1
89346	0	0	Bin 1
125131	17	13	Bin 3
38574	2	15	Bin 2
43069	60	54	Bin 3
7448	0	0	Bin 1
46013	0	4	Bin 1
39795	0	0	Bin 1
4131	0	0	Bin 1
111519	24	7	Bin 3
89671	0	0	Bin 1
121716	0	0	Bin 1
74329	1	0	Bin 2
24876	0	11	Bin 1
95149	0	0	Bin 1
8511	29	3	Bin 3
119515	0	8	Bin 1
89965	271	275	Bin 5
108035	0	0	Bin 1
52931	8	0	Bin 2
66290	1	0	Bin 2
61862	0	0	Bin 1
73090	0	0	Bin 1
118191	47	35	Bin 3
118825	15	0	Bin 3
13295	0	0	Bin 1
19981	30	22	Bin 3
21992	0	0	Bin 1
101236	4	0	Bin 2
17707	0	0	Bin 1
125570	3	0	Bin 2

Arrival Delay in Minutes in Equi-width:

id	Arrival Delay in M	Arrival Delay in Minutes [Binned]
49833	0	Bin 1
22532	0	Bin 1
111275	5	Bin 1
74985	0	Bin 1
75332	7	Bin 1
106685	5	Bin 1
86460	0	Bin 1
81846	24	Bin 1
102610	5	Bin 1
114955	0	Bin 1
105654	28	Bin 1
101667	43	Bin 1
49295	0	Bin 1
85252	0	Bin 1
118594	30	Bin 1
114930	0	Bin 1
117832	10	Bin 1
8915	14	Bin 1
74178	77	Bin 1
62993	0	Bin 1
64239	22	Bin 1
22078	39	Bin 1
95784	13	Bin 1
84358	0	Bin 1
89346	0	Bin 1
125131	13	Bin 1
38574	15	Bin 1
43069	54	Bin 1
7448	0	Bin 1
46013	4	Bin 1
39795	0	Bin 1
4131	0	Bin 1
111519	7	Bin 1
89671	0	Bin 1
121716	0	Bin 1
74329	0	Bin 1
24876	11	Bin 1
95149	0	Bin 1
8511	3	Bin 1
119515	8	Bin 1
89965	275	Bin 1
108035	0	Bin 1
52931	0	Bin 1
66290	0	Bin 1
61862	0	Bin 1
73090	0	Bin 1
118191	35	Bin 1
118825	0	Bin 1
13295	0	Bin 1
19981	22	Bin 1
21992	0	Bin 1
101236	0	Bin 1
17707	0	Bin 1
125570	0	Bin 1

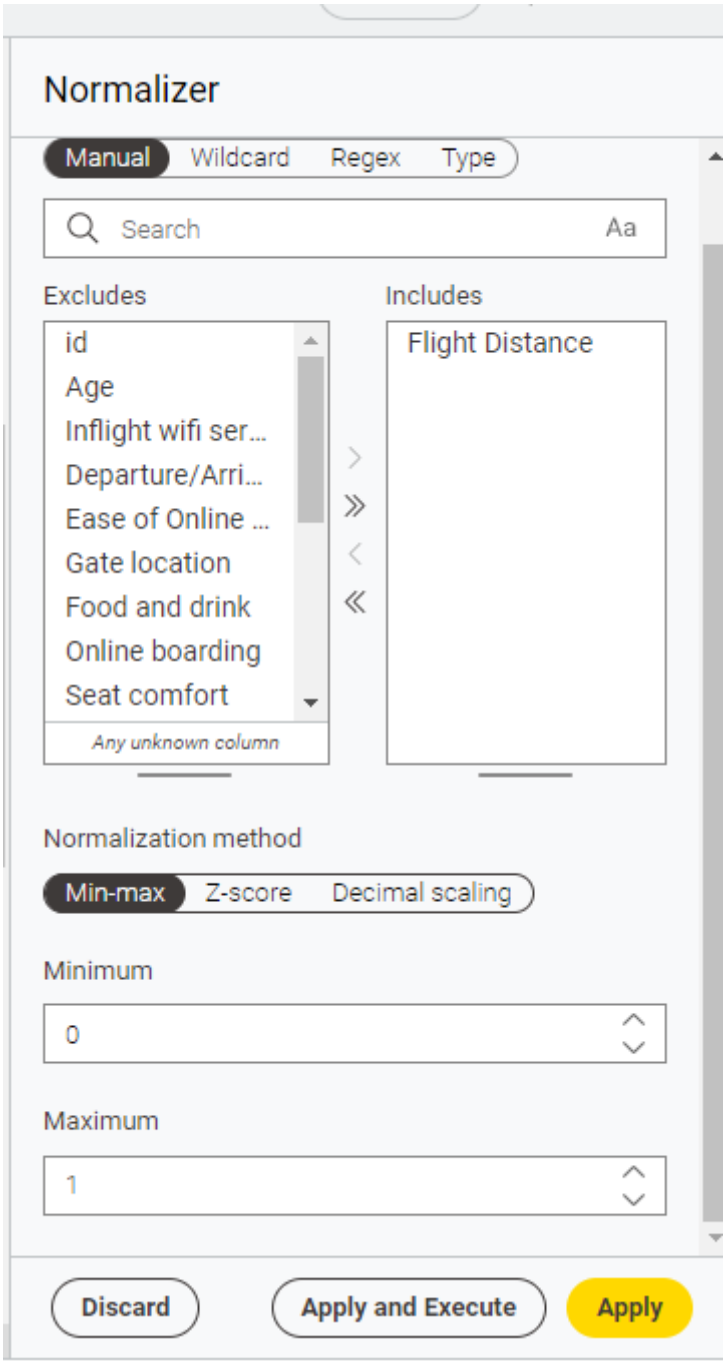
Arrival Delay in Minutes in Equi-depth:

A	B	C	D	E
id	Arrival Del	Arrival Delay in Minutes [Binned]		
49833	0	Bin 1		
22532	0	Bin 1		
111275	5	Bin 2		
74985	0	Bin 1		
75332	7	Bin 2		
106685	5	Bin 2		
86460	0	Bin 1		
81846	24	Bin 3		
102610	5	Bin 2		
114955	0	Bin 1		
105654	28	Bin 3		
101667	43	Bin 3		
49295	0	Bin 1		
85252	0	Bin 1		
118594	30	Bin 3		
114930	0	Bin 1		
117832	10	Bin 2		
8915	14	Bin 2		
74178	77	Bin 3		
62993	0	Bin 1		
64239	22	Bin 3		
22078	39	Bin 3		
95784	13	Bin 2		
84358	0	Bin 1		
89346	0	Bin 1		
125131	13	Bin 2		
38574	15	Bin 3		
43069	54	Bin 3		
7448	0	Bin 1		
46013	4	Bin 2		
39795	0	Bin 1		
4131	0	Bin 1		
111519	7	Bin 2		
89671	0	Bin 1		
121716	0	Bin 1		
74329	0	Bin 1		
24876	11	Bin 2		
95149	0	Bin 1		
8511	3	Bin 2		
119515	8	Bin 2		
89965	275	Bin 5		
108035	0	Bin 1		
52931	0	Bin 1		
66290	0	Bin 1		
61862	0	Bin 1		
73090	0	Bin 1		
118191	35	Bin 3		
118825	0	Bin 1		
13295	0	Bin 1		
19981	22	Bin 3		
21992	0	Bin 1		
101236	0	Bin 1		
17707	0	Bin 1		
125570	0	Bin 1		

2. Ⅰ. min-max normalization to transform the values onto the range [0.0-1.0].

Min-max normalization: Scales data to the [0,1] interval, suitable for visualization and neural network modeling.

Example of setting in Knime:

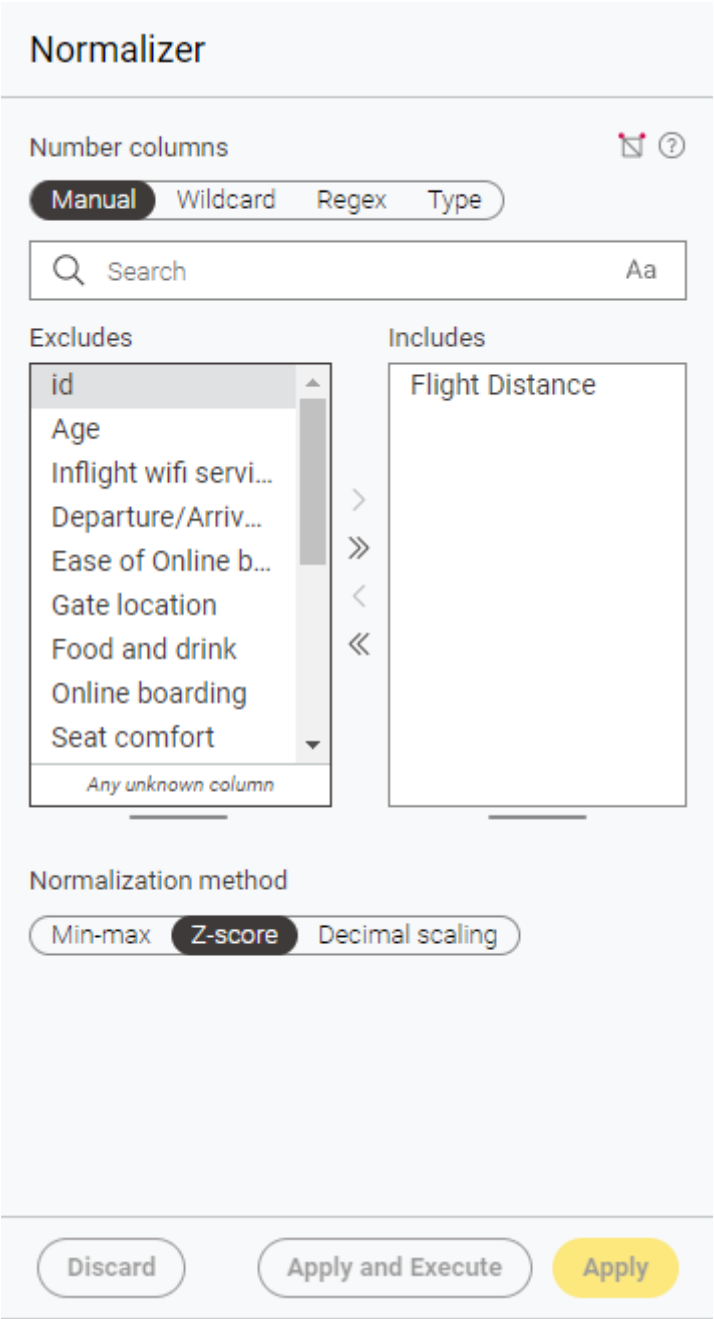


id	Flight Distance
49833	0.55311
22532	0.022617
111275	0.08643
74985	0.493538
75332	0.210016
106685	0.084814
86460	0.147617
81846	0.294023
102610	0.071688
114955	0.075323
105654	0.163772
101667	0.419023
49295	0.055937
85252	0.132876
118594	0.068457
114930	0.077948
117832	0.059976
8915	0.127221
74178	0.123183
62993	0.189015
64239	0.278877
22078	0.055129
95784	0.085218
84358	0.116115
89346	0.314216
125131	0.06664
38574	0.493538
43069	0.046042
7448	0.099152
46013	0.487278
39795	0.048667
4131	0.103595
111519	0.072698
89671	0.718901
121716	0.131664
74329	0.289378
24876	0.06563
95149	0.032512
8511	0.167811
119515	0.175283
89965	0.257472
108035	0.055533
52931	0.130856
66290	0.165792
61862	0.256866
73090	0.212843
118191	0.285339
118825	0.05937
13295	0.149435
19981	0.112682
21992	0.095315
101236	0.314216
17707	0.056543
125570	0.039378

Ⅱ. z-score normalization to transform the values.

After using Z-score standardization, the Flight Distance value is converted to a distribution with a mean of approximately 0 and a standard deviation of 1, which is beneficial for subsequent K-Means clustering.

Example of setting in Knime:

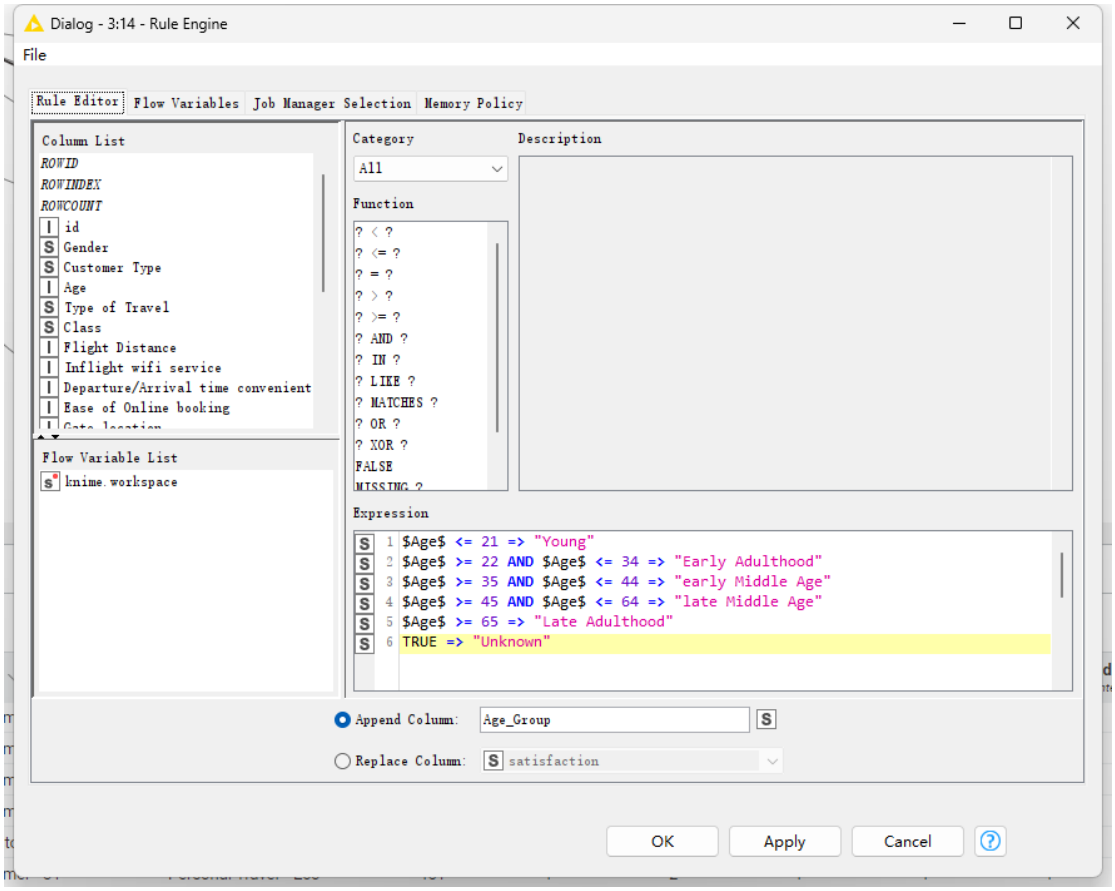


id	Flight Distance
49833	1.608552
22532	-1.04512
111275	-0.72591
74985	1.310556
75332	-0.1077
106685	-0.734
86460	-0.41984
81846	0.312524
102610	-0.79966
114955	-0.78147
105654	-0.33903
101667	0.937809
49295	-0.87845
85252	-0.49358
118594	-0.81582
114930	-0.76834
117832	-0.85824
8915	-0.52186
74178	-0.54207
62993	-0.21276
64239	0.236762
22078	-0.88249
95784	-0.73198
84358	-0.57742
89346	0.413539
125131	-0.82491
38574	1.310556
43069	-0.92795
7448	-0.66227
46013	1.279242
39795	-0.91481
4131	-0.64005
111519	-0.7946
89671	2.437888
121716	-0.49964
74329	0.28929
24876	-0.82996
95149	-0.99563
8511	-0.31882
119515	-0.28145
89965	0.129686
108035	-0.88047
52931	-0.50368
66290	-0.32892
61862	0.126656
73090	-0.09356
118191	0.269087
118825	-0.86128
13295	-0.41075
19981	-0.59459
21992	-0.68147
101236	0.413539
17707	-0.87542
125570	-0.96128

3. Discretise the age attribute into the following categories

   I used a Rule Engine node to convert the continuous variable Age into a categorical variable, Age-Group. This allows for easier comparison and analysis of passengers of different age groups (for example, differences in satisfaction and cabin preferences among age groups).

Example of setting in Knime:

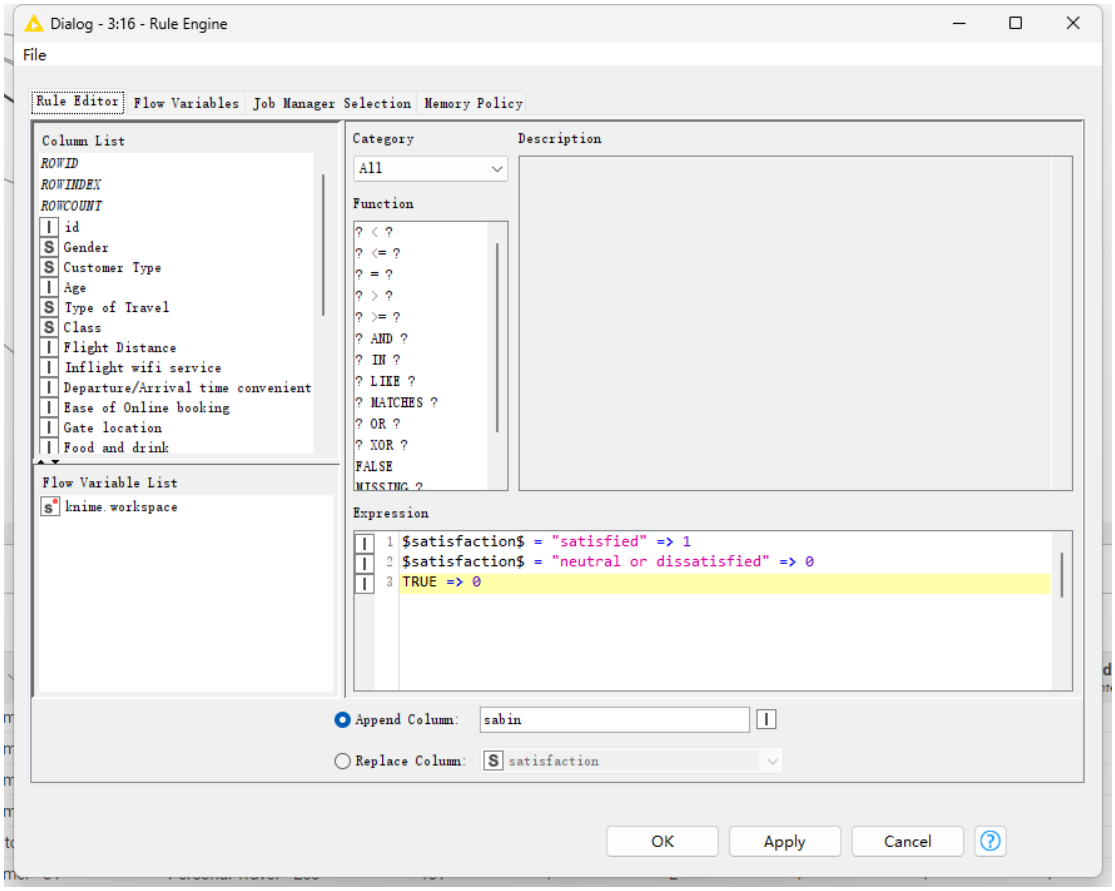


id	Age	Age_Group
49833	41	early Middle Age
22532	29	Early Adulthood
111275	59	late Middle Age
74985	62	late Middle Age
75332	29	Early Adulthood
106685	31	Early Adulthood
86460	9	Young
81846	48	late Middle Age
102610	38	early Middle Age
114955	33	Early Adulthood
105654	26	Early Adulthood
101667	19	Young
49295	45	late Middle Age
85252	50	late Middle Age
118594	34	Early Adulthood
114930	22	Early Adulthood
117832	33	Early Adulthood
8915	25	Early Adulthood
74178	21	Young
62993	45	late Middle Age
64239	11	Young
22078	17	Young
95784	62	late Middle Age
84358	49	late Middle Age
89346	29	Early Adulthood
125131	7	Young
38574	47	late Middle Age
43069	12	Young
7448	23	Early Adulthood
46013	44	early Middle Age
39795	33	Early Adulthood
4131	43	early Middle Age
111519	44	early Middle Age
89671	31	Early Adulthood
121716	46	late Middle Age
74329	19	Young
24876	17	Young
95149	36	early Middle Age
8511	38	early Middle Age
119515	40	early Middle Age
89965	52	late Middle Age
108035	50	late Middle Age
52931	20	Young
66290	22	Early Adulthood
61862	18	Young
73090	17	Young
118191	18	Young
118825	45	late Middle Age
13295	16	Young
19981	52	late Middle Age
21992	18	Young
101236	31	Early Adulthood
17707	38	early Middle Age
125570	37	early Middle Age

4. Binarise the satisfaction variable [with values "0" or "1"]

   I used a Rule Engine node to convert the text field satisfaction into a numeric variable sabin. This facilitates subsequent cluster analysis, correlation calculation, and visualization.

Example of setting in Knime:



id	satisfaction	sabin
49833	neutral or dissatisfied	0
22532	neutral or dissatisfied	0
111275	neutral or dissatisfied	0
74985	neutral or dissatisfied	0
75332	neutral or dissatisfied	0
106685	neutral or dissatisfied	0
86460	neutral or dissatisfied	0
81846	neutral or dissatisfied	0
102610	neutral or dissatisfied	0
114955	neutral or dissatisfied	0
105654	neutral or dissatisfied	0
101667	neutral or dissatisfied	0
49295	neutral or dissatisfied	0
85252	neutral or dissatisfied	0
118594	neutral or dissatisfied	0
114930	neutral or dissatisfied	0
117832	neutral or dissatisfied	0
8915	neutral or dissatisfied	0
74178	neutral or dissatisfied	0
62993	neutral or dissatisfied	0
64239	neutral or dissatisfied	0
22078	neutral or dissatisfied	0
95784	neutral or dissatisfied	0
84358	neutral or dissatisfied	0
89346	neutral or dissatisfied	0
125131	neutral or dissatisfied	0
38574	neutral or dissatisfied	0
43069	neutral or dissatisfied	0
7448	neutral or dissatisfied	0
46013	neutral or dissatisfied	0
39795	neutral or dissatisfied	0
4131	neutral or dissatisfied	0
111519	neutral or dissatisfied	0
89671	neutral or dissatisfied	0
121716	neutral or dissatisfied	0
74329	neutral or dissatisfied	0
24876	neutral or dissatisfied	0
95149	neutral or dissatisfied	0
8511	neutral or dissatisfied	0
119515	neutral or dissatisfied	0
89965	neutral or dissatisfied	0
108035	neutral or dissatisfied	0
52931	neutral or dissatisfied	0
66290	neutral or dissatisfied	0
61862	neutral or dissatisfied	0
73090	neutral or dissatisfied	0
118191	neutral or dissatisfied	0
118825	neutral or dissatisfied	0
13295	neutral or dissatisfied	0
19981	neutral or dissatisfied	0
21992	neutral or dissatisfied	0
101236	neutral or dissatisfied	0
17707	neutral or dissatisfied	0
125570	neutral or dissatisfied	0

1C. Summary

Analysis of the airline passenger dataset revealed several key findings:

Demographic Distribution: Passenger ages ranged from 7 to 85, with an average age of approximately 39. The overall age distribution was broad and nearly symmetrical, indicating that the airline serves a highly diverse customer base.

Flight Distance Characteristics: The average flight distance was approximately 1,177 kilometers, but the distribution exhibited a significant right-skewed distribution. Most flights were short-haul (<1,000 kilometers), while a small number of long-haul flights (>3,000 kilometers) were associated with greater delay fluctuations.

Service Ratings: Inflight Wi-Fi received the lowest satisfaction rating (mean 2.74/5), while baggage handling (3.64/5) and inflight service (3.63/5) received the highest ratings. This suggests that while basic services performed well overall, digital services (such as Wi-Fi) remained a major source of passenger dissatisfaction.

Flight Delays: Both departure and arrival delays exhibited an extremely skewed distribution. While most flights were on time or only slightly delayed, a few extreme flights experienced delays exceeding 1,500 minutes, significantly distorting the overall distribution. These significant delays require special attention from management.

Passenger Group Clustering: K-Means clustering (K=3) identified three main groups:

Young passengers (average age 22) had the lowest satisfaction rate (approximately 30%);

Middle-aged passengers (average age 40) had higher satisfaction rates (approximately 50%);

Older passengers (average age 57) had similar satisfaction rates to the middle-aged group (approximately 49%).

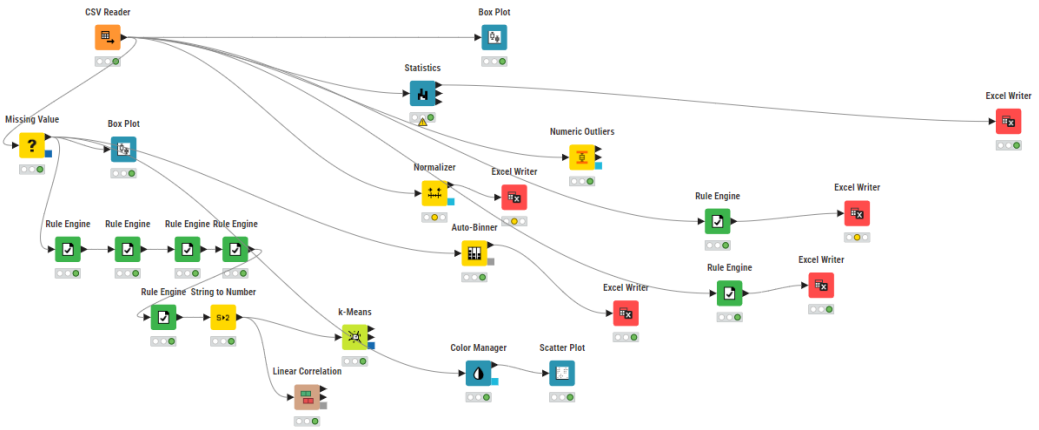
This suggests that younger passengers may have higher expectations for services (such as Wi-Fi and entertainment facilities), leading to lower satisfaction.

Variable Relationships:

Business class passengers are generally older, while economy class passengers tend to be younger.

Age showed a weak correlation with most service ratings, with only slight positive correlations for online check-in ($r \approx 0.21$) and seat comfort ($r \approx 0.17$).

The scatter plot shows that long-haul flights are less frequent but more prone to significant delays, while short-haul flights are more numerous, with delays primarily concentrated in the 0–100 minute range.

Overall, my analysis highlights several key insights: younger passengers have lower satisfaction, inflight Wi-Fi service is a persistent weakness, and long-haul flights have a higher risk of significant delays. These areas warrant further research by the analytics department to support service improvements and optimize resource allocation.

Knime workfram: